

Aarogya Sahayak × Swytchcode

Governed Agent Tool Execution, Clinical Idempotency & Indic Voice Infrastructure

Architecture: Multi-Agent Intelligence + Swytchcode Runtime • **Deployment:** Render Singapore (FastAPI) + Vercel (PWA)

1. Executive Problem Statement & Core Value

In rural healthcare, when an AI assistant identifies critical danger signs—such as a mother with **severe pre-eclampsia (BP 165/105 mmHg with blurred vision)**—allowing an autonomous LLM to invoke external APIs directly is hazardous. Unchecked LLM tool calling leads to **hallucinated parameters, duplicate ambulance dispatches over flaky 2G/3G networks, and credential leaks**.

Aarogya Sahayak integrates Swytchcode as its enterprise-grade AI execution & governance layer. The AI reasoning engine never sees or holds external API credentials. Every medical alert, Sarvam Indic voice synthesis, and facility query executes through Swytchcode with **sliding-window idempotency, strict pre-execution schema validation, and real-time observability**.

■ ■ Guaranteed Idempotency Emergency dispatches deduplicated via 5-min sliding window SHA-256 tokens. Zero duplicate ambulances.	■ Zero-Token Security LLM reasoning never touches production API secrets. All credentials isolated in Swytchcode vault.
■ ■ Sarvam Indic Voice Proxy Marathi & Hindi speech (Saaras & Bulbul) governed with 3s latency budgets and phonetic fallback.	■ Live Observability Every tool call streams live to app.swytchcode.com with latency, status 200 OK, and schema logs.

2. System Architecture & Governed Data Flow

3. Architectural Evolution: Before vs. With Swytchcode

Capability	Before Swytchcode (Legacy)	With Swytchcode (Governed Runtime)
Tool Security	Raw API keys exposed in app memory; vulnerable to prompt extraction.	Zero-Token Isolation: LLM emits intent only. Keys locked in Swytchcode vault.
Emergency Idempotency	Flaky 2G/3G network drops caused clients to retry -> 3 to 5 duplicate ambulance dispatches.	Deterministic Deduplication: SHA-256 idempotency suppresses duplicate alerts within 5 min.
Clinical Validation	Ad-hoc try/catch blocks. Malformed blood pressure or oxygen values could hit webhooks.	Pre-Execution Guardrail: Pydantic schema validation rejects invalid clinical payloads before network dispatch.
Sarvam Indic Voice	Direct REST calls to Sarvam. Socket drops on rural towers caused app to crash.	Governed Voice Proxy: Swytchcode manages 3s timeout budgets, retries, and local phonetic fallback.
Live Telemetry	Raw terminal stdout logs. Zero visual proof for judges or health auditors.	Live Dashboard: Real-time event telemetry stream at app.swytchcode.com.

4. The 3 Governed Healthcare Tools in Aarogya Sahayak

1. dispatch_emergency_asha_alert Trigger: Severe pre-eclampsia, cardiac, or pediatric danger signs. Action: Dispatches urgent clinical escalation notification to assigned ASHA & doctor queue. Swytchcode Governance: Ingress schema validation (BP, SpO2, weeks), SHA-256 idempotency deduplication, zero PII exposure.
2. sarvam_indic_voice_gateway Trigger: Citizen speaks in Marathi (mr-IN) or Hindi (hi-IN). Action: Routes audio through Sarvam AI (Saaras STT & Bulbul TTS). Swytchcode Governance: Enforces 3,000ms latency budget, language allowlist, and graceful phonetic fallback.
3. query_health_facility_registry Trigger: Patient or ASHA worker searches for nearest facility with ICU, NICU, or 24x7 emergency. Action: Queries verified Ayushman Bharat PM-JAY empanelled facilities. Swytchcode Governance: Read-only execution boundary; blocks any unauthorized database write mutations.

5. Step-by-Step Live Proof & Verification Guide (For Judges)

Use these commands on stage to demonstrate verified live execution to the judges:

TEST 1: Swytchcode Runtime Health & Registered Tools

```
curl -X GET 'https://your-backend.onrender.com/api/swytchcode/status'
```

Expected Output: `{'status': 'LIVE_CONNECTED', 'workspace_alias': 'calm-meadow-c150', 'tools_registered': ['dispatch_emergency_asha_alert', 'sarvam_indic_voice_gateway', ...]}`

TEST 2: Live Governed Emergency Triage Dispatch

```
curl -X POST 'https://your-backend.onrender.com/api/swytchcode/execute-tool' -H 'Content-Type: application/json' -d '{"tool_name": "dispatch_emergency_asha_alert", "priority": "CRITICAL", "clinical_condition": "Severe pre-eclampsia: BP 165/105"}'
```

Expected Output: `{'status': 'DISPATCHED', 'trace_id': 'SWY-EMG-C0E4A3D2', 'latency_ms': 135.2, 'idempotency_enforced': true, 'dashboard_audit_url': 'https://app.swytchcode.com/dashboard/overview'}`

TEST 3: The Idempotency Test (Duplicate Suppression Defense)

Run the exact same curl command a second time within 5 minutes:

Expected Output: `{'status': 'ALREADY_DISPATCHED_IDEMPOTENT', 'idempotency_hit': true, 'message': 'Duplicate emergency alert suppressed by Swytchcode idempotency engine.'}`

6. Architecture Diagram Visual Prompt (For Diagrams / Slides)

Paste the prompt below into **Eraser.io**, **Napkin AI**, or **Mermaid Live Editor** to render high-resolution diagram graphics:

```
graph TD
  A[Citizen Mobile PWA / Voice] -->|HTTPS Audio/Symptoms| B(FastAPI Backend)
  B --> C{Deterministic Rule Engine}
  C -->|Emergency Detected| D[PII Masking Engine]
  D -->|Sanitized Intent| E[Swytchcode Governance Runtime]
  E -->|Idempotent Webhook| F[ASHA & Doctor Emergency Queue]
  F -->|Governed Proxy| G[Sarvam AI Indic Voice Saaras/Bulbul]
  G -->|Live Telemetry| H[Swytchcode Dashboard app.swytchcode.com]
```

■■ 2-Minute Winning Pitch Script for Judges:

"Namaste judges. In rural India, when an AI assistant detects that a pregnant mother has a critical blood pressure of 165/100, allowing an LLM to directly call external APIs is dangerous. Models hallucinate parameters, rural 3G network drops cause duplicate ambulance dispatches, and credentials can leak.

*We integrated **Swytchcode** as our enterprise AI tool execution & governance layer. Every single action—from **Sarvam AI Indic voice translation** in Marathi and Hindi to **emergency ASHA dispatches**—is governed by Swytchcode.*

*As you can see right here on our live **Swytchcode Dashboard at app.swytchcode.com**, the triage alert was executed with status 200 OK, latency 135ms, strict schema validation, zero-token security, and guaranteed idempotency. Swytchcode makes AI in healthcare safe, deterministic, and ready for 1.4 billion citizens!"*